Causal Dissection of Scientific Agents Breaks from Evidence to Action

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Abstract

Experimental success does not reveal whether a scientific agent learned the right model. The same endpoint can follow from an inherited prior, productive search, phenomenological interpolation or evidence-driven revision. We use ChemWorld as a controlled causal probe: the executable world, interface, resources and stochastic identity remain fixed while the agent's initial model is made opaque, aligned or misspecified at entity, parametric or structural loci. Across 135 public agent sessions, these interventions redirected search and prediction errors fell in every arm and locus, yet the prespecified selective-correction criterion failed at all three loci. Under matched structural counterevidence, mean normalized errors converged from 0.2255/0.2736/0.3392 to 0.0074/0.0060/0.0071, but 0/5 misspecified summaries recovered the registered 1.75 power law and all five adopted saturation or endpoint models. All 135 typed laws executed, yet 84 preserved less predictive information than the agents' final explicit predictions. In fresh multi-task action selection, only 11/42 eligible readouts selected the best unseen plan, and thresholded law adequacy did not determine a correct action. Finally, a complete-ranking control passed 7/8 fresh 96-query units while selecting Top-1 in only 1/8, whereas a fresh 320-query unit failed the same rank gate but selected Top-1 with zero regret. Experimental optimization, numerical revision, structural identification, executable compression, decision transfer and evaluator validity are therefore distinct transitions rather than one scientific-agency score.

1. Introduction

Autonomous experimental agents are increasingly judged by whether they discover high-performing molecules or reactions. Yet an endpoint benchmark cannot distinguish whether an agent inherited a fortunate prior, navigated a productive search corridor, adopted a phenomenological interpolation or genuinely revised its scientific understanding from evidence. We call this ambiguity endpoint underdetermination: the outcome alone cannot identify which route produced it. The question is not whether an AI scientist finds a good experiment, but what that experiment changes in its model of the world.

This distinction is becoming consequential as language-model agents plan syntheses, call chemistry tools, operate instruments and participate in self-driving laboratory workflows [4, 5, 8, 20–22]. Interactive environments likewise test repeated cycles of hypothesis formation, intervention and inference [1, 9, 11, 13, 26, 27]. Yet apparent scientific success remains difficult to interpret because pretrained knowledge, prompt-provided information, experiment selection, endpoint optimization and verbal explanation are usually observed together. A useful outcome does not reveal whether evidence changed where an agent searched, improved its counterfactual predictions, weakened a contradicted model, produced an executable relation or supported a decision under conditions it had not encountered.

The unresolved problem is one of identification. Physical autonomous laboratories establish consequential experimental competence, while virtual discovery environments make repeated studies scalable. Neither setting generally provides the matched counterfactual needed here: the same executable world encountered by agents that receive equally explicit but differently correct initial models. Without that comparison, a correct prior can masquerade as rapid discovery, a wrong prior can improve an endpoint by encouraging useful exploration without ever being rejected, and a plausible verbal law can remain inconsistent with the agent's predictions or actions. Scientific correction therefore cannot be inferred from endpoint score or self-report alone.

We address this problem by treating the agent's initial world model as an experimental variable. ChemWorld thereby functions as a controlled causal probe of scientific agency: the external chemistry, action space, observations, resources and stochastic identity remain fixed while the agent-facing description is made opaque, aligned with the world or systematically misspecified at one declared locus [18]. The intervention can target entity identity, dynamical parameters or process structure. One persistent agent then conducts a shared-resource campaign, and evaluator-owned counterfactuals measure how the intervention propagates through experiment selection, prediction, explicit model revision and terminal action. The ChemWorld foundation study establishes the executable substrate and its replay properties; those platform qualifications are not reused here as evidence of agent capability.

We applied this design to a fixed DeepSeek-V4-Flash experimental-agent configuration across 45 matched task–

world clusters and 135 prospective campaigns. Fixed checkpoints bound the participant's beliefs to independently evaluated predictions and typed executable summaries. Matched-evidence sessions separate failure to acquire diagnostic evidence from failure to revise a model after seeing it. Blind incumbent replay tests reproducibility of a committed observed action, while a separate multi-task assay reveals eight complete, previously unseen ActionPlans only after autonomous exploration and measures terminal ranking without hidden workflow defaults. A five-condition follow-up was designed to compare no evidence, yoked evidence, autonomous exploration, a transferred learned law and a provider-free oracle law, but its oracle control failed the frozen rank gate in a fresh formal world before any participant session began. Expanding the oracle fitting grid from 96 to 320 queries repaired exposed construction failures but did not restore prospective complete-ranking validity. A frozen reanalysis then showed that rank-gate success and correct action were not interchangeable. These outcomes distinguish experimental search, predictive learning, selective model correction, executable-summary fidelity, unseen action selection and evaluator validity rather than collapsing them into one score.

The result is not one success or failure but four breaks in a putative evidence-to-action chain. First, initial models scaffolded search without inducing selective repair of the wrong model. Second, matched counterevidence produced near-exact numerical revision without recovery of the governing structure. Third, executable summaries often lost information present in explicit predictions. Fourth, the final break appeared at two levels: at the agent level, thresholded law adequacy did not determine unseen-plan selection; at the evaluator level, complete-ranking qualification did not determine action validity. These downstream dissociations are not causal mediation estimates, but together they show that scientific agency behaves as a sequence of separable transitions rather than a scalar capability. By holding the world fixed while intervening on the starting model, the framework shifts evaluation from whether an agent finds a good experiment to identifying which scientific transformations its evidence actually supports.

2. Related work

2.1 Autonomous experimentation and laboratory agents

Autonomous chemistry systems combine planning, analysis and physical execution. Coscientist, ChemCrow, A-Lab and instrument-facing agents demonstrate tool use, synthesis planning, materials search and closed-loop experimentation on real equipment [4, 5, 7, 8, 20, 21]. These systems establish that AI agents can participate in consequential experimental workflows. Their main evidential strength is physical validity; their practical constraint for the present question is the cost of repeatedly constructing matched alternative worlds in which only the correctness of a prior changes.

2.2 Virtual chemistry and process environments

Summit and Olympus support repeatable reaction optimization and experimental-design benchmarks, while PC-Gym exposes nonlinear process-control problems [3, 10, 12]. ChemGymRL provides modular interactive benches for reaction, extraction, distillation and characterization, enabling reinforcement-learning agents to act within a virtual chemistry laboratory [2]. MADE extends closed-loop discovery benchmarks to budget-constrained materials settings [15]. These systems make experimentation cheaper, safer and more repeatable than physical execution, but they are generally used to compare policies or optimize endpoints under one task definition. The present study instead uses world programmability to hold the executable task fixed while intervening on the agent's initial scientific model.

2.3 Interactive scientific-discovery benchmarks

DiscoveryWorld, BoxingGym, SciGym and SciExplorer organize scientific problems around repeated cycles of hypothesis formation, intervention and inference [9, 11, 13, 17]. PhysGym is the closest controlled-prior neighbor: it varies the level of prior knowledge available to agents exploring interactive physics systems [6]. NewtonBench and DiscoverPhysics instead construct counterfactual or non-canonical physical worlds and ask agents to recover their hidden laws through experimentation [24, 27]. CausaLab separates task success from recovery of the underlying structural causal mechanism, while ReplaySCM evaluates executable mechanisms by held-out interventional replay [1, 26]. SciDisco turns process-verifiable discovery environments into intermediate training signals [25].

Another line develops discovery algorithms rather than diagnostic interventions. A probabilistic formulation treats language-model proposals and revisions as inference over mechanistic simulator models [23]. LLM-AutoSciLab couples hypothesis proposal, informative experiment selection and iterative mechanism refinement, whereas the Model Discovery Agent combines language-model structure proposals with sequential Monte Carlo, simulation-based inference and value-of-information design [14, 16]. These methods ask how to make mechanistic discovery more accurate or data efficient. The present study asks a complementary causal question about a fixed persistent agent: when the executable world, action space and resources are held constant, how does changing the correctness of its initial world model alter search, prediction, correction, executable-law recovery and unseen-action selection?

2.4 The unresolved identification problem

Large-scale behavioral evidence already suggests that successful scientific workflows need not be accompanied by evidence-sensitive, self-correcting reasoning [19]. However, an observational comparison across systems cannot by itself identify which capability transition produced the divergence. Three ambiguities remain when endpoint score, belief statement and scientific understanding are not separated. First, a correct prior can make an agent look like a

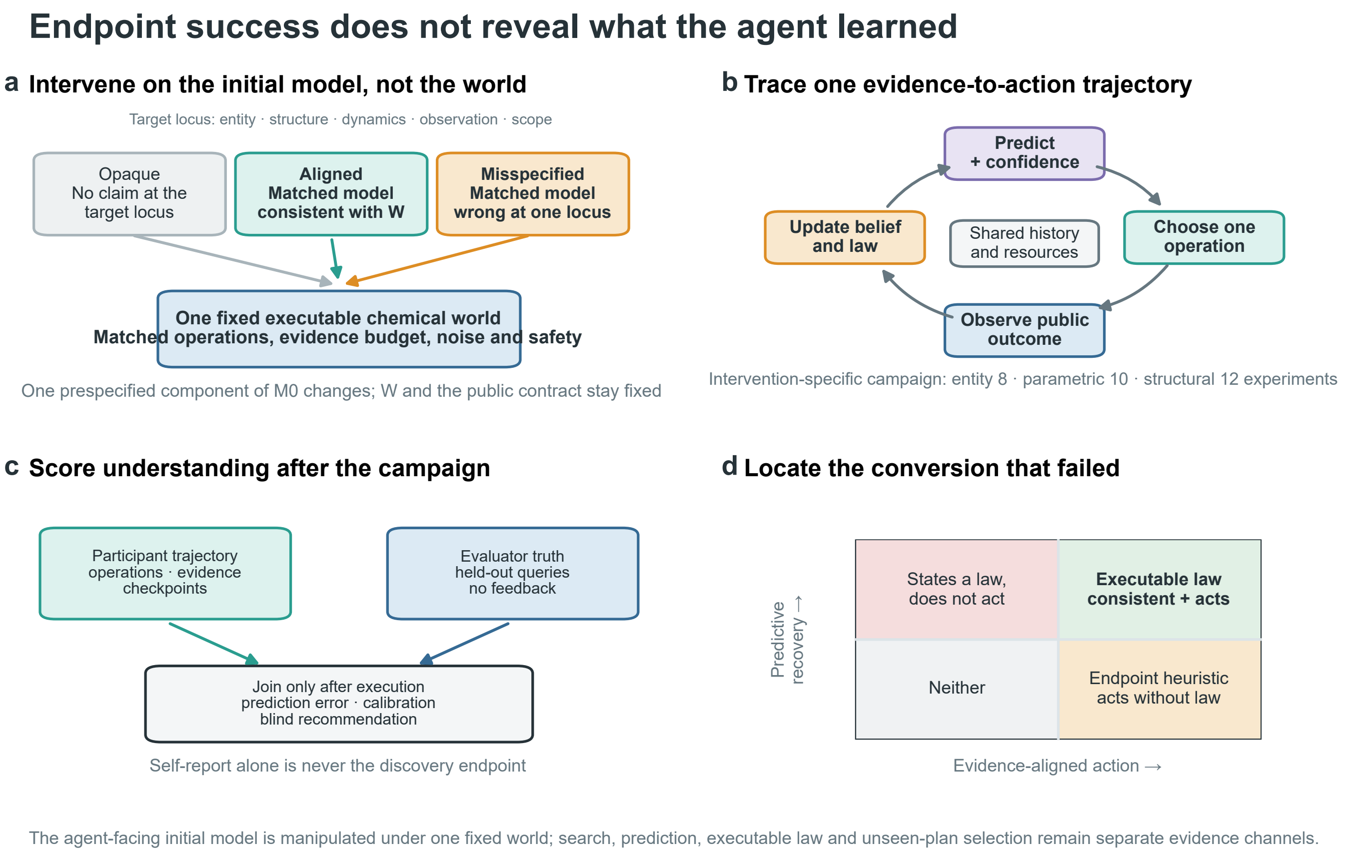

Figure 1. Endpoint success does not reveal what the agent learned. **a,** The current entity-level instantiation uses opaque, aligned and misindexed dossiers in the same fixed executable world; the same intervention logic can target structural, parametric or observation-model assumptions. **b,** One persistent session repeatedly predicts, selects an operation, observes the public outcome and updates its belief and executable law summary across a shared-resource campaign. **c,** Participant trajectories and evaluator-owned held-out truth remain separate until the campaign ends; prediction error, calibration and blind recommendation outcomes are scored afterward. **d,** Predictive recovery and evidence-aligned unseen-plan selection define four distinguishable phenotypes. Endpoint success or a correct statement alone does not identify which conversion succeeded.

rapid discoverer even if it merely confirms supplied information. Second, a wrong prior can improve an endpoint by encouraging useful exploration without ever being rejected. Third, a verbal law summary can be correct while the agent's subsequent predictions or actions remain inconsistent with it. A matched correctness intervention on the initial model, typed checkpoints and evaluator-owned tests are needed to distinguish these cases. Unlike counterfactual-world benchmarks, the hidden law does not change within a matched comparison here; unlike prior-availability studies, aligned and misspecified arms receive equally explicit models and differ at the targeted scientific locus.

3. Conceptual framework

3.1 Fixed world, programmable initial world model

Each matched comparison begins with one executable world containing a fixed evaluator-owned law. Write the world as $W = (\mathcal{E}, G, \Theta, O, C)$: entities, causal/mechanistic structure, parameters and dynamics, observation mapping and the authoritative public contract. The agent instead begins with $M_0 = (\widehat{\mathcal{E}}, \widehat{G}, \widehat{\Theta}, \widehat{O}, \widehat{S})$, where $\widehat{S}$ represents assumptions about scope, modularity and compositional applicability. The public task, action space, actual observation channels, resource card, safety rules and bound stochastic identity are held constant. We intervene only on one declared component of M_0 before the first experiment. Changing W or the public contract would create a different task; changing one component of M_0 creates a controlled epistemic intervention within the same task.

The programmable intervention space has four scientific layers and one non-intervention boundary.

Within any selected layer, intervention quality follows the same logic:

- **Opaque:** no additional task-specific claim is supplied for the manipulated layer.
- **Aligned:** incomplete information is directionally consistent with the fixed world.
- **Misspecified:** an equally detailed and equally confident model is wrong at the manipulated layer.

The current entity/ontology implementation realizes the misspecified condition through a misindexed dossier: the

Table 1. Programmable layers of the agent's initial world model. The external executable world and public contract remain fixed within every matched comparison.

Initial-model layer	What may be aligned or wrong	Role in this paper
Entity / ontology	identity–property mappings, entity classes and property bundles	exploratory and prospective tests
Structural / mechanistic	causal topology, active process modules and dominant-pathway assumptions	separately prespecified test
Parametric	coefficient signs, thresholds, orderings and plausible ranges	separately prespecified test
Observation model	instrument mapping, reliability, bias and noise assumptions	secondary diagnostic probe
Scope / compositionality	applicability domains, invariant modules and transfer boundaries	context-reset artifact-portability study
Contract / resource boundary	budget, safety, action permissions and actual observation interface	authoritative and fixed; never treated as a manipulable prior

same property bundles, fields, values, wording and confidence language are retained, but the bundles are permuted across material identifiers. Structural, parametric and observation-model interventions require their own matched encodings and identifiability validation; they cannot be declared equivalent to material misindexing after observing the result. Scope/compositionality is tested only when a learned artifact enters a fresh target context. Experimental evidence remains explicitly authoritative in all conditions. A wrong initial model is therefore not a trick question about obedience; it tests whether the agent seeks and uses evidence that can override a plausible scientific representation.

3.2 Operation, experiment and campaign

An operation is one submitted physical or measurement action. A complete experiment begins with a new batch and ends with a committed final assay or an allowed discard. A discovery campaign comprises multiple complete experiments under one fixed world, one persistent agent session and one shared resource ledger. Physical batch state resets between experiments, but the public history, agent context, hidden law and remaining resources persist.

This distinction matters because repeated operations and experiments inside one campaign are nested observations, not independent scientific samples. The independent analysis unit is the matched task-by-world cluster.

3.3 Five separable outcomes and one capability chain

We distinguish five outcomes that are often collapsed into a single score, while recording the intermediate transitions that connect them.

1. **Endpoint optimization:** whether the campaign identifies a high-quality experimental outcome.
2. **Predictive recovery:** whether held-out counterfactual prediction error decreases.
3. **Prior correction:** whether evidence selectively improves the wrong-prior condition without degrading the correct-prior condition.
4. **Executable-law consistency:** whether the final typed summary executes on prespecified held-out coordinates and preserves the quality of the agent's conditional predictions.
5. **Unseen-plan selection:** whether the terminal agent state supports ranking and selecting previously unseen, fully specified executable plans rather than merely retrieving an observed incumbent.

The process-level chain is therefore

$$\text{initial world model} \rightarrow \text{experiment selection} \rightarrow \text{evidence acquisition}$$
$$\rightarrow \text{prediction / belief update} \rightarrow \text{executable law} \rightarrow \text{unseen action selection} \rightarrow \text{artifact portability}$$

The paper reports transition losses rather than a composite intelligence score: evidence-to-prediction loss, prediction-to-law loss, law-to-action inconsistency and action-to-artifact-portability loss. This makes it possible to identify where a capability fails without treating a successful endpoint as proof that every upstream step was correct.

An agent can succeed on any subset. In particular, endpoint success without predictive and transfer validity is classified as local optimization rather than law discovery.

4. Study design

4.1 Chemical-world cohort and intervention studies

The study is layer-stratified. Its entity/ontology backbone spans electrochemical conversion, reaction followed by crystallization, reaction followed by distillation, phase-partition discovery and safety-constrained reaction. Five independently selected public worlds per task yield 25 task-by-world clusters and 75 participant cells across opaque, aligned and misspecified arms. Parametric/dynamical and structural/mechanistic blocks each contain two independently validated task families and five worlds per task, adding 30 participant cells per locus. Exploratory, validation and prospective world instances remain disjoint. A sealed private cohort is an optional stronger study and was not executed in the current paper.

This design uses ChemWorld's programmability to manipulate different components of M_0, but does not turn the paper into a full factorial benchmark. Every block changes one locus, has its own identifiability criterion and retains its own denominator:

1. **Initial-model-conditioned free discovery.** Entity interventions span five task families; parametric and structural interventions each span two validated task families. A cross-locus claim requires evidence from all three prespecified blocks; an entity-only result remains entity-specific.
2. **Matched-evidence falsification.** A cloned-world secondary probe presents the same contradictory evidence to each arm, separating failure to seek evidence

from failure to update after seeing it. The analysis includes the unaffected parametric block and a corrected structural phase-process block. An earlier structural run was excluded because its evaluator truth source omitted the prespecified world intervention. All matched-evidence sessions are independent and excluded from the free-discovery denominator.

3. **Executable law and action.** Typed law summaries and held-out predictions test the transition from conditional belief to executable relation. Blind incumbent replay tests whether a committed recommendation can reproduce observed value, while a separate longitudinal open-action assay tests whether the agent can rank previously unseen, fully specified ActionPlans; no verbal statement alone counts as discovery.
4. **Future artifact portability (not executed).** After source-world learning, raw evidence, prose summaries or executable laws could be transferred to a context-reset agent in a new combination. This future study is not part of the present evidence, and within-family replication remains separate from compositional transfer.

Observation/measurement interventions are reserved as a separate boundary probe. They require two-task identifiability and an exploratory three-arm study and are not included in the present denominator. Scope/compositional assumptions are tested only in the future portability study. This preserves a complete conceptual intervention space without claiming that every programmable coordinate has already been executed.

Environment validation and participant outcomes form separate evidence layers. Environment tests establish that the hidden relations are coherent, identifiable and executable through the public measurement surface. They do not show that an agent discovers those relations.

4.2 Persistent experimental agent

Each cell is controlled by one persistent agent process across one campaign. After every public outcome, the participant chooses the next operation through the host-owned laboratory tool. The host validates schemas, executes transactions, updates resources and protects private state, but does not select or repair scientific actions.

Campaign length is intervention-specific: entity studies use eight complete experiments with checkpoints after 0, 2, 4, 6 and 8 experiments; parametric studies use ten with checkpoints after 0, 2, 4, 7 and 10; structural studies use twelve with checkpoints after 0, 3, 6, 9 and 12. These pattern-owned counts and their finite resource cards were fixed before participant execution. A checkpoint records the agent's assessment of initial-model reliability at the manipulated layer, predictions, uncertainty, evidence references, executable law summary and next experimental intent. Checkpoints do not create additional independent sessions.

4.3 Evaluator-owned evidence

The participant predicts four prespecified counterfactual queries per entity checkpoint and 16 per parametric or structural checkpoint. The evaluator executes each unique task-world query set independently; truth is shared across prior arms and checkpoints and is never returned to the participant. The primary prediction error is the mean normalized absolute error across prespecified query-metric pairs. The held-out evaluation is complete: 420/420 truth executions produced 1,620 query–metric truth values, and all 675 checkpoints were scored without additional model calls.

The evaluator also executes each final typed law summary on the same prespecified query coordinates. This produces a separate cell-level record of schema validity, complete query-metric executability, truth-normalized error, error change relative to the pre-evidence and effective-final checkpoints, and consistency with the participant's final explicit predictions. These public measures are descriptive and are reported as executable-law consistency/compression; they do not establish reusable-law status. A reusable or transferable law would require independent coordinates or a context-reset transfer test.

After the campaign, the participant commits one completed experiment as its final recommendation. The evaluator then performs paired blind replay of the observed incumbent and the committed recommendation. These executions use separate resources and do not enter participant-operation or independent-sample denominators.

4.4 Longitudinal open-action assay

The longitudinal action assay addresses a different estimand from incumbent replay. One persistent agent first completes 12 autonomous experiments and mechanism checkpoints after 0, 3, 6, 9 and 12 experiments. Only after the final checkpoint is committed does the host reveal eight previously unseen ActionPlans. Each public plan includes its ordered operations, all submitted parameters, initial-state assumptions, measurement positions, terminal assay and omitted-operation declarations; candidate outcomes, evaluator ranks and other-arm evidence remain hidden. The evaluator executes exactly the public plan and verifies identity among the disclosed, truth-evaluated and executed plans.

The formal multi-task matrix contains three task families, five worlds per task and three initial-model arms, giving 45 scheduled cells. Each cell contains 12 autonomous experiments and five checkpoints; 42 cells were eligible for action metrics and three crystallization failures remained in the scheduled denominator. The primary action endpoint is within-world regret of the selected plan; selected rank, Top-1, complete ranking and law adequacy are reported separately. This matrix is analyzed separately from the prospective locus tests, and no arm-level inference is made when a world lacks a complete triplet.

A separate causal follow-up prespecified five information conditions within each task–world–prior stratum: no evidence, stepwise evidence yoked from an autonomous donor,

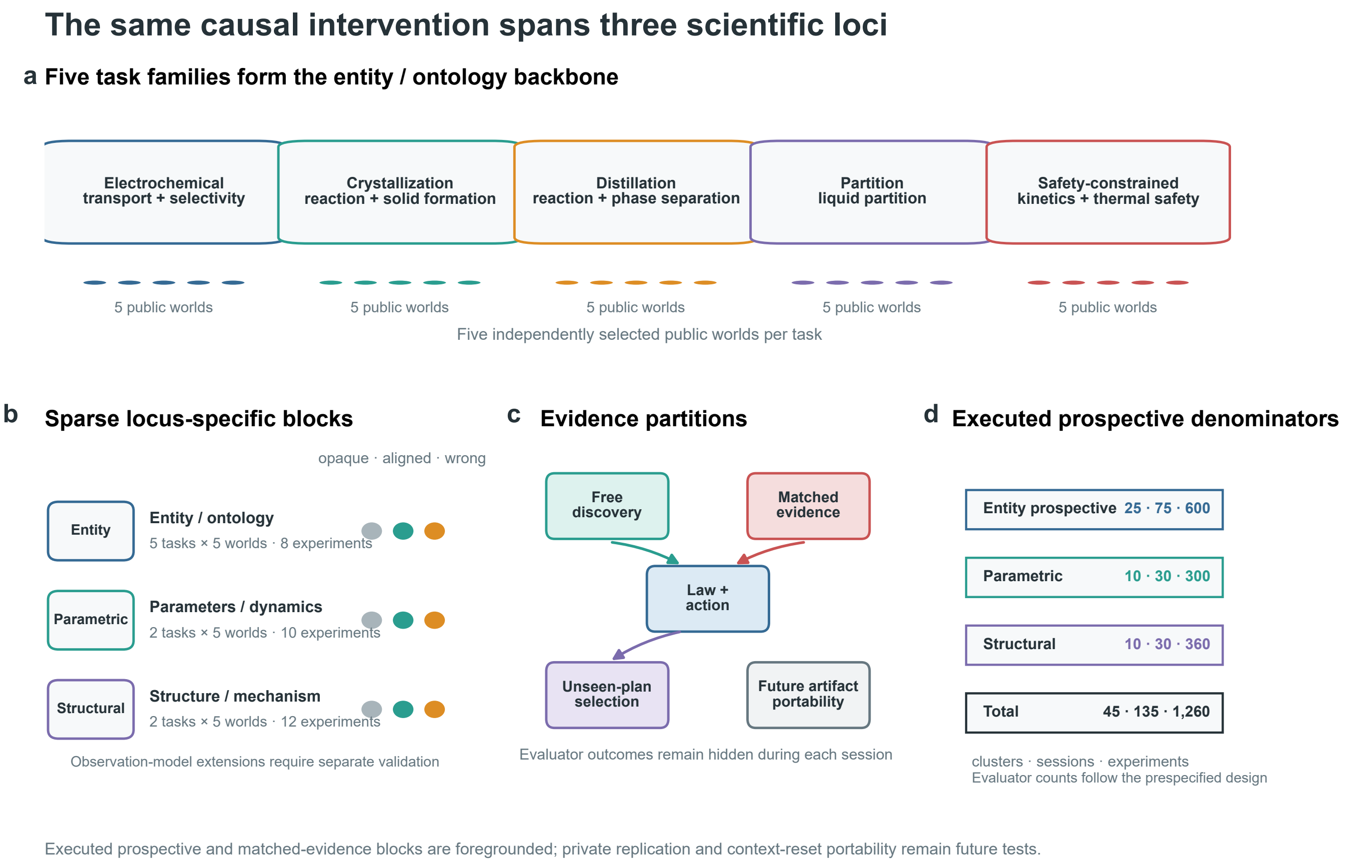

Figure 2. Layer-stratified study architecture. a, The entity/ontology backbone uses five task families and five independently selected public worlds per task. **b,** Every matched task–world cluster contains opaque, aligned and misspecified initial models for one declared locus. Campaign length and checkpoints are owned by the locus pattern rather than forced into one universal four-experiment limit. **c,** Executed free-discovery, matched-evidence, law and unseen-plan assays retain separate sessions, evidence and denominators; context-reset portability is shown in grey as future work. **d,** The prospective entity, parametric and structural blocks total 45 task–world clusters, 135 sessions and 1,260 planned experiments. The diagram is a design map, not outcome evidence.

autonomous exploration, the donor's learned law in a fresh context and a provider-free oracle law in a fresh context. Across three tasks, five formal worlds and three priors, the intended denominator was 225 sessions; only the 45 autonomous donors would have executed 12 experiments each. Before participant execution, every task–world required an outcome-disjoint oracle law to reproduce the eight-candidate ordering with Spearman rank correlation at least 0.80. Failure rejected the control design rather than authorizing world replacement or revealing an outcome table.

We subsequently evaluated the control rather than weakening this rule. A denser construction used 64 global queries and 256 candidate-neighborhood queries while keeping the fitted model family fixed. Exposed construction worlds and new prospective worlds remained separate. A final zero-execution diagnostic re-read all completed 96- and 320-query unit versions and compared the frozen Spearman gate with Top-1 selection, near-optimality and regret. It added no participant, provider or physical execution and did not alter any earlier stop decision.

4.5 Hypotheses and estimands

Let $E^{(\ell)}_{a,k}$ denote held-out prediction error for initial-model arm a, checkpoint k and intervention locus ℓ. Each block tests selective evidence-driven correction:

$$C_\ell = (E^{(\ell)}_{\text{misspecified,pre}} - E^{(\ell)}_{\text{misspecified,final}}) - (E^{(\ell)}_{\text{aligned,pre}} - E^{(\ell)}_{\text{aligned,final}}).$$

For the entity locus, the misspecified arm is instantiated by a prespecified material permutation and $C_\ell = C_E$ is the confirmatory contrast. Success requires the lower confidence bound for the locus-specific contrast to exceed zero, the wrong-prior condition to improve, and the aligned condition not to deteriorate beyond a prespecified tolerance. Correct-prior utility, wrong-prior vulnerability and knowledge-to-action translation form a hierarchical secondary family. A cross-locus conclusion requires concordant, separately reported entity, parametric and structural results; standardized effects may be synthesized hierarchically, but raw contrasts are not pooled as if their intervention semantics were identical. Observation-model results remain a distinct boundary analysis unless evaluated in a separate prespecified study. Endpoint, calibration, behavior, law-summary, terminal-action, resource and safety outcomes are reported as separate channels rather than one leaderboard score.

Failed scientific cells remain in the denominator and are not replaced. A right-censored cell carries its last valid checkpoint forward; a missing final prediction receives zero primary improvement. Only a pure infrastructure failure without a persisted trajectory may resume under the prespecified attempt cap.

5. Exploratory studies establish the identification problem

Development studies were used to discover the inferential traps that the prospective design had to remove. Across configuration-separated entity-level matrices, explicit initial information changed where the agent searched and sometimes improved the best observed endpoint, but the ordering was not aligned, opaque and then misspecified. In the retained five-task development cohort, all 75 trajectories replayed exactly and 69 met the prespecified eligibility criteria. Held-out predictions usually improved, yet the misspecified arm did not improve more than the aligned arm on average. Executable summaries frequently lost information present in checkpoint predictions, and blind recommendations almost always retrieved an observed incumbent. These results are protocol-development evidence, not part of the prospective denominator and not a cross-system comparison.

The same ambiguity appeared at the parameter level. In a one-world preliminary study, the misspecified agent tested inside its supplied potential window, obtained a zero score, sharply reduced its stated confidence in that model and moved the next experiment outside the window. This was behavioral rejection of a supplied model, but the agent still failed to recover the best finite-budget policy. Rejecting a wrong prior, improving prediction and recovering a useful law were therefore not interchangeable outcomes.

These exploratory observations motivated three safeguards in the main study: matched worlds with equally explicit aligned and misspecified models; evaluator-owned counterfactual predictions rather than endpoint score or verbal suspicion; and separate assays for executable-law fidelity and terminal selection among unseen plans. The prospective results below begin from this stricter identification problem. Exploratory configurations remain archived with their exact denominators and failures but do not compete with the main causal narrative.

6. Causal dissection of the evidence-to-law chain

6.1 The prospective cohort closes the experimental and evaluator denominators

The prospective analysis combines 120 unaffected sessions with a complete 15-session replacement of the structural crystallization block after correction of its resource contract. All **135/135** scheduled sessions produced final records. The participant completed **1,243/1,260** planned experiments, and **121/135** sessions met the prespecified operational eligibility criteria. The denominator contains 1,269 closed batch lifecycles: 1,243 ended in a final assay and 26 were discarded. No dynamic physical failure occurred. Thirteen operations were rejected by the finite laboratory resource ledger, and 84 attempted operations were not committed. Failures and right-censoring remain in their assigned denominators.

Every session submitted all five checkpoints, giving **675/675** typed belief snapshots, 6,300 prespecified counterfactual predictions and 24,300 query–metric values. The evaluator independently completed **420/420** truth queries and scored every checkpoint without additional participant calls. The cohort therefore permits a continuous analysis from the starting model through search, prediction and executable summary rather than a comparison of endpoint scores alone.

6.2 Priors scaffold search without imposing one performance ordering

A retrospective manipulation analysis shows that the initial-model intervention reached both the reported belief state and the first experimental trajectory. Pre-evidence prediction errors differed across arms, although the aligned arm was not uniformly best. More directly, the first complete recipe differed between aligned and misspecified cells in **45/45** matched task–world clusters, between opaque and aligned cells in **45/45**, and between opaque and misspecified cells in **44/45**. Because the design did not include repeated same-arm sessions, exact-recipe divergence is a trajectory-sensitivity check rather than a new confirmatory estimand; provider stochasticity cannot be separated from intervention sensitivity at this resolution.

The ensuing campaigns were not simple copies of the first proposal. Across cells, **91.2%** of completed experiments used a unique recipe, **84.4%** of session optima occurred after the midpoint and **32.6%** occurred in the final completed experiment. Intermediate measurement was task appropriate rather than globally absent: 666/1,269 closed lifecycles included a non-final measurement, comprising 872 instrument uses. The intervention changed the trajectory, and the agent continued to collect and use public outcomes.

Those trajectory changes produced three recurring endpoint patterns. In entity-level partition, aligned information produced a durable advantage over the misspecified arm: +0.106 on the first experiment and +0.200 at the best endpoint, both concordant in 5/5 worlds. In structural crystallization, the aligned model gave a +0.141 first-experiment head start in 5/5 worlds, but the best-endpoint gap narrowed to +0.055 as the initially disadvantaged arm explored. In structural partition, both aligned and misspecified descriptions outperformed opaque identifiers while differing little from one another, consistent with shared search scaffolding rather than correct-model utility. The remaining task–locus groups were heterogeneous. A correct initial model was therefore neither a universal advantage nor a stable endpoint ordering.

6.3 Predictive learning does not selectively repair a wrong starting model

All three intervention loci showed mean pre-to-final prediction-error reductions in every arm. For the entity locus, reductions were 0.111, 0.097 and 0.097 for opaque, aligned and misspecified cells; for the parametric locus they were 0.090, 0.033 and 0.065; and for the structural locus they were 0.219, 0.228 and 0.221. The agent acquired predictive information during free discovery.

The prespecified causal estimand was stricter. Evidence should improve the misspecified arm more than the aligned arm while preserving aligned performance. This selective-correction criterion failed at all three loci. Failure-aware contrasts were **-0.214** for entity ($p = 0.990$), **+0.033** for parametric ($p = 0.079$) and **-0.224** for structural ($p = 1.000$). The positive parametric direction is a replication hypothesis, not a passed locus. Structural crystallization and partition also pointed in opposite directions, defeating the required cross-task structural decision. General predictive learning and selective correction of a wrong starting model are thus empirically different transitions.

6.4 Matched counterevidence drives numerical convergence but not structural-law recovery

Free discovery cannot by itself distinguish failure to seek diagnostic evidence from failure to use evidence once obtained. The matched-evidence assay holds the evidence itself fixed. In the parametric block, all five misspecified summaries rejected the supplied high-potential direction and recovered the peak-and-collapse response. The free-discovery loss therefore contains an evidence-acquisition component: when the decisive response profile was supplied, the agent could recognize it.

The corrected structural assay produced a sharper dissociation. After all three arms received the same direct phase-process evidence, mean normalized errors fell from 0.2255, 0.2736 and 0.3392 to **0.0074, 0.0060 and 0.0071** for opaque, aligned and misspecified cells. The misspecified-minus-aligned update-gain contrast was +0.0645, but only 3/5 worlds were positive (exact one-sided sign-flip $p = 0.125$; descriptive 95% interval $[-0.0557, 0.1848]$). This is numerical convergence under matched evidence, not a confirmatory arm effect.

Structural identification nevertheless failed. **0/5** misspecified public summaries recovered the prespecified 1.75 power law, only **1/5** explicitly rejected the supplied linear form, and **5/5** shifted to a saturation or endpoint model. Direct evidence was sufficient for accurate local phenomenological interpolation but not for identifying the governing relation. Evidence acquisition, numerical belief revision and structural-law identification are therefore separate transitions. The intervention is causal at the starting prior; this downstream localization is a diagnostic dissociation, not a causal mediation analysis.

6.5 Executable laws often lose information present in predictive belief

All **135/135** final typed laws executed on their prespecified continuous coordinates, but executability did not preserve the information in the agent's conditional predictions. Mean law MAE was 0.237. Relative to the effective final predictions, executable laws were better in 50 cells, equal in one and worse in 84; mean law-minus-final error was +0.069. Even in the structural locus, where pre-to-law improvement was largest, law error remained higher than final explicit-prediction error on average. The typed interface solved syntax and coverage, not faithful model compression.

A same-domain schema-capacity control localized that loss. For every cell, we fitted the best legal identity-link typed law directly to the agent's complete final prediction vector and executed it through the production parser. The full registered basis reproduced the prediction state in **135/135** cells (mean MAE 4.25×10^{-13}), whereas the participant's submitted law differed from the same prediction state by 0.1539 MAE. Holding the participant's term set fixed reduced that gap to 0.0114 MAE (58/135 near-exact cells), and leave-one-query-out fitting yielded 0.0788 MAE. Thus the interface can carry the observed predictive state; most information loss arose during the agent's distillation into a sparse law, not from schema capacity. Because the full-basis fit uses coordinates from the same evaluation domain, it is a capacity control rather than evidence of global mechanistic identification.

Paired blind replay tested a still narrower claim: whether the final commitment reproduced the value of an observed incumbent. It completed **726/726** executions for 121 evaluable cells; the 14 failed or right-censored cells remained as unstarted. Recommendations were better, equivalent and worse than the incumbent in **1/119/1** cells. This demonstrates reliable incumbent retrieval, not selection beyond the observed campaign. A separate assay is required for unseen plans.

The checkpoint interface is part of this evaluated system. Although all checkpoints were eventually recovered, 888 typed-checkpoint submissions were rejected before acceptance. This burden did not remove prediction payloads, but it increased context and recovery work. Results therefore apply to the complete agent–tool configuration rather than the base language model alone.

7. Scientific laws, unseen actions and evaluator validity

7.1 Complete plan semantics remove a hidden-workflow explanation

Blind incumbent replay asks whether a recommendation can reproduce an observed batch; it does not test decisions outside the campaign history. We therefore used a second assay in which the agent first explores freely for 12 experiments and only then ranks eight unseen plans. Every candidate is a complete ActionPlan, including ordered

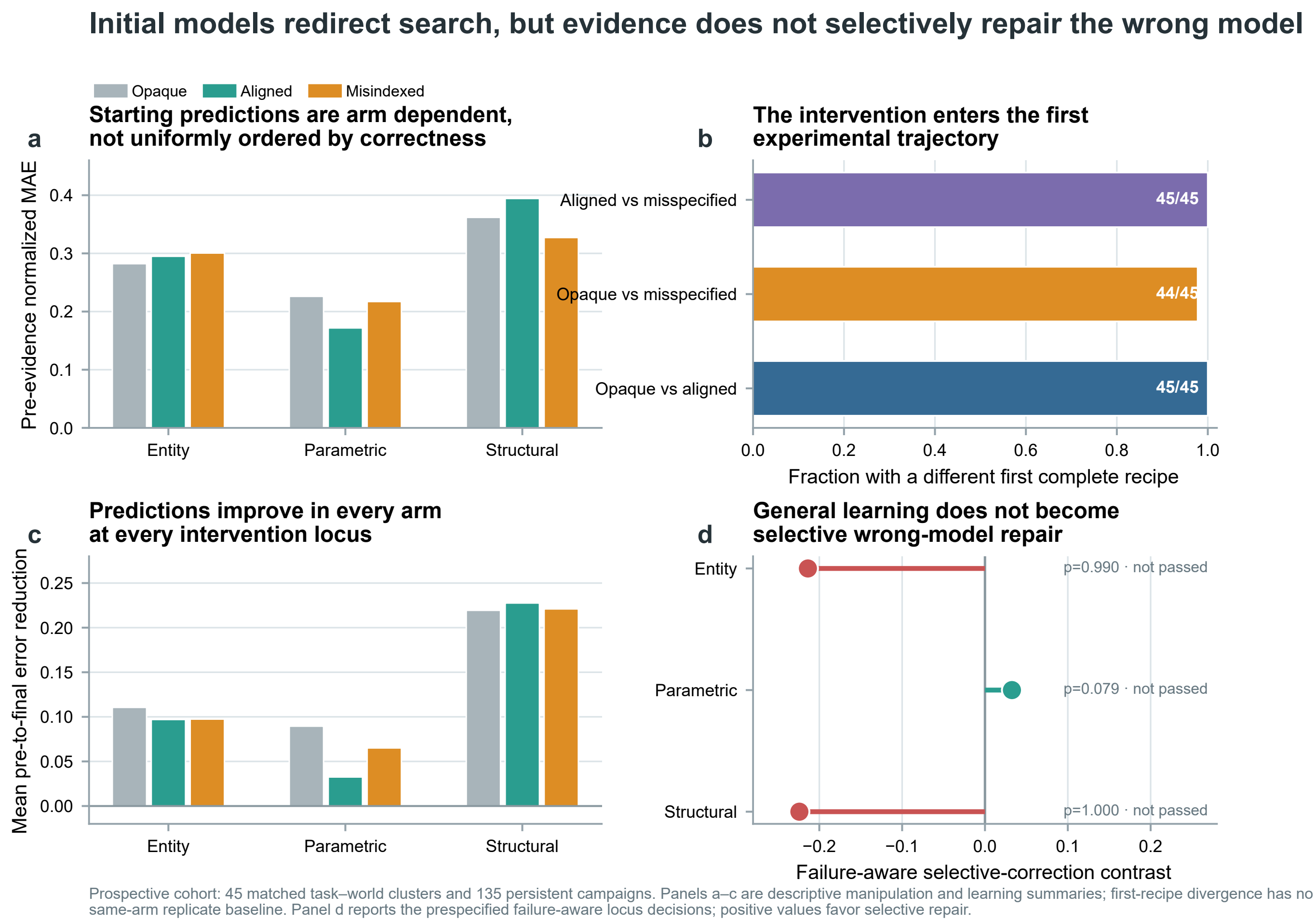

Figure 3. Initial models redirect search, but evidence does not selectively repair the wrong model. a, Mean pre-evidence normalized prediction error by locus and initial-model arm; correctness does not impose a uniform starting-error ordering. **b,** Fraction of 45 matched task–world clusters in which the first complete experimental recipe differs between each pair of arms. This retrospective manipulation check has no repeated same-arm baseline. **c,** Mean pre-to-final prediction-error reduction. Every arm improves at every locus. **d,** Prespecified failure-aware selective-correction contrasts and locus p values. Positive values favor greater repair in the misspecified arm; no locus passed.

operations, submitted parameters, measurement positions and terminal assay. Public, evaluator-truth and executed plans were verified as identical, and no evaluator-owned default could silently alter a workflow. Candidate outcomes and ranks remained hidden. An incorrect ranking therefore cannot be attributed to undisclosed execution semantics.

7.2 Agent-level law adequacy does not determine unseen action selection

The formal matrix produced final records for **45/45** scheduled cells across three tasks, five worlds and three initial-model arms. Independent evaluation completed **240/240** truth executions and **240/240** exact replays. **42/45** cells were uncontaminated and eligible for action metrics. The three excluded cells were all crystallization cells: two exhausted agent-selected resources or process options and one was right-censored by interruption. All three remain in the scheduled denominator.

Across eligible cells, **11/42** terminal readouts selected the true Top-1 plan; mean selected rank was 3.31/8 and mean normalized regret was 0.297. The uniform-random rank of 4.5 is a geometric reference, not a causal baseline: the design contains neither a no-evidence action arm nor a pre-exploration ranking from the same agent. The study therefore describes post-campaign selection competence but does not estimate how much exploration or a recovered law caused it.

The joint mechanism–action result exposes the boundary more sharply. **30/42** cells had an inadequate law and wrong action, **11/42** had an inadequate law but correct action, **1/42** had an adequate law but wrong action, and **0/42** combined an adequate law with a correct action. The single law-adequate/wrong-action cell is a counterexample to logical guarantee, not a population estimate of law sufficiency; correct action also occurred without thresholded law adequacy.

The conclusion did not depend on the 0.10 adequacy cutoff. Across all 42 eligible cells, law MAE had weak pooled associations with selected rank (Spearman $\rho = -0.073$, task–world cluster-bootstrap 95% interval $[-0.380, 0.256]$)

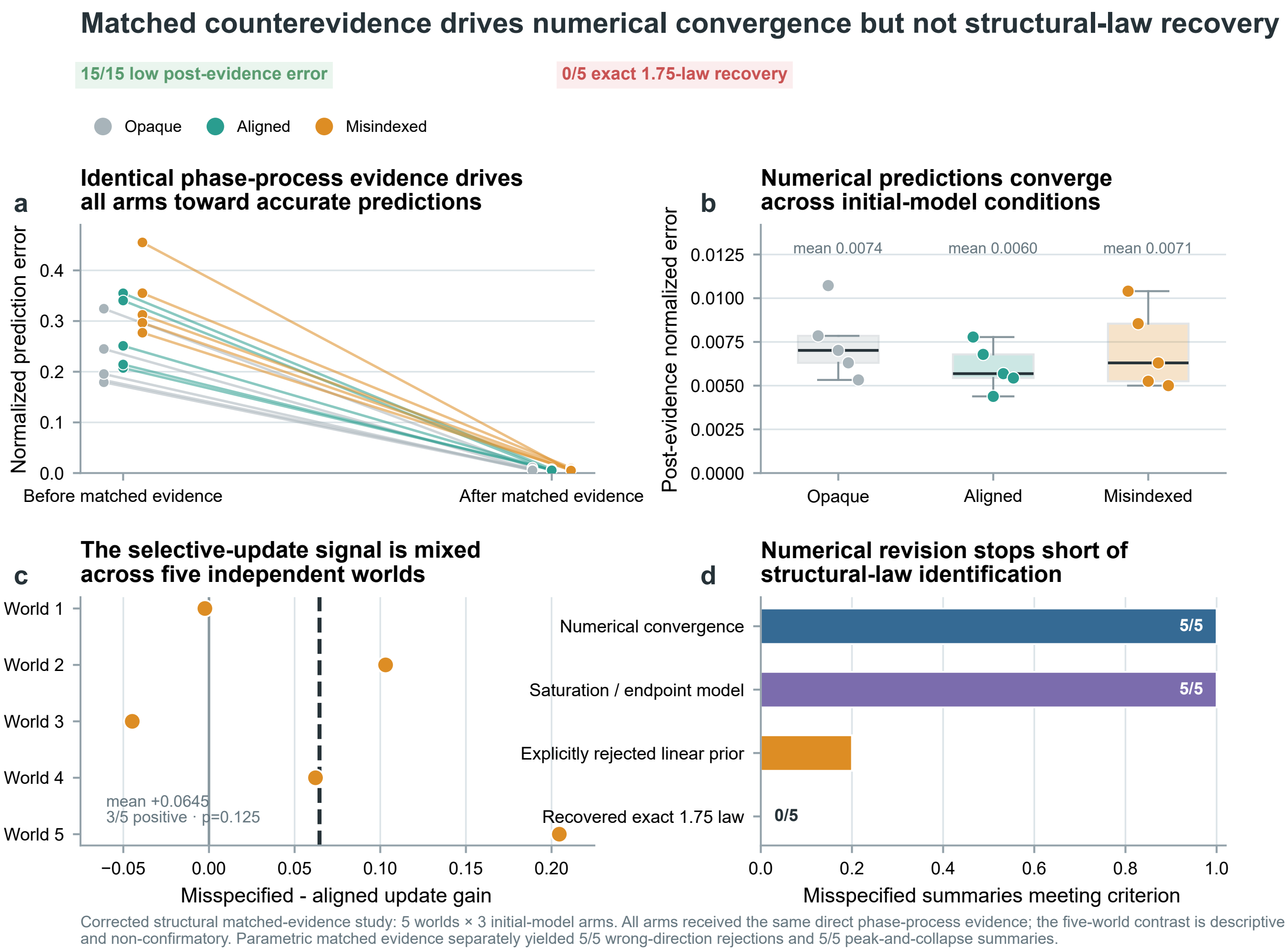

Figure 4. Matched counterevidence drives numerical convergence but not structural-law recovery. a, Pre-to-post structural prediction errors for all 15 corrected matched-evidence cells; every arm receives identical direct phase-process evidence. **b,** Post-evidence errors converge near zero in opaque, aligned and misspecified cells. **c,** Misspecified-minus-aligned update-gain contrasts across five independent worlds; the dashed line is the mean and the five-world result is non-confirmatory. **d,** Public-summary recovery among the five misspecified cells. Numerical convergence and saturation-style empirical models are universal, whereas explicit rejection of the linear prior is rare and exact 1.75-law recovery is absent.

Table 2. Formal multi-task unseen-plan matrix. Rank and regret summaries use the 42 eligible cells; all 45 scheduled cells remain in the denominator and the three crystallization failures are retained. Lower rank and regret are better.

Arm	Scheduled	Eligible	Mean rank	Top-1	Mean normalized regret
Opaque	15	14	3.14	5	0.2742
Aligned	15	14	3.36	3	0.2958
Misspecified	15	14	3.43	3	0.3222

and normalized regret ($\rho = -0.133$, $[-0.452, 0.217]$). Task-stratified rank associations reversed direction: $+0.524$ for electrochemical conversion, -0.592 for reaction safety and -0.007 for crystallization. Sweeping the law-MAE threshold from 0.05 to 0.30 increased the adequate-law subset from 1 to 34 cells, yet its correct-action count rose only from 0 to 9; at no threshold did adequacy become sufficient for a correct action. The binary four-way table is therefore one readable slice of a continuous, task-dependent non-monotonic relation rather than an artifact of a single cutoff.

Performance varied substantially by task. Electrochemical conversion reached Top-1 in 4/15 cells with mean rank 3.60, reaction safety in 4/15 with mean rank 2.00, and crystallization in 3/12 with mean rank 4.58. Every pairwise arm-rank contrast changed sign under some leave-one-cluster-out omission. Arm means are therefore descriptive, not causal. The bounded result is that complete public action semantics and autonomous exploration did not yield uniformly reliable ranking of unseen plans.

7.3 The repair is sensitivity evidence, not a replacement cell

An independent aligned-arm crystallization repair completed all 12 experiments and produced a terminal readout, but the agent first proposed an infeasible seeding operation

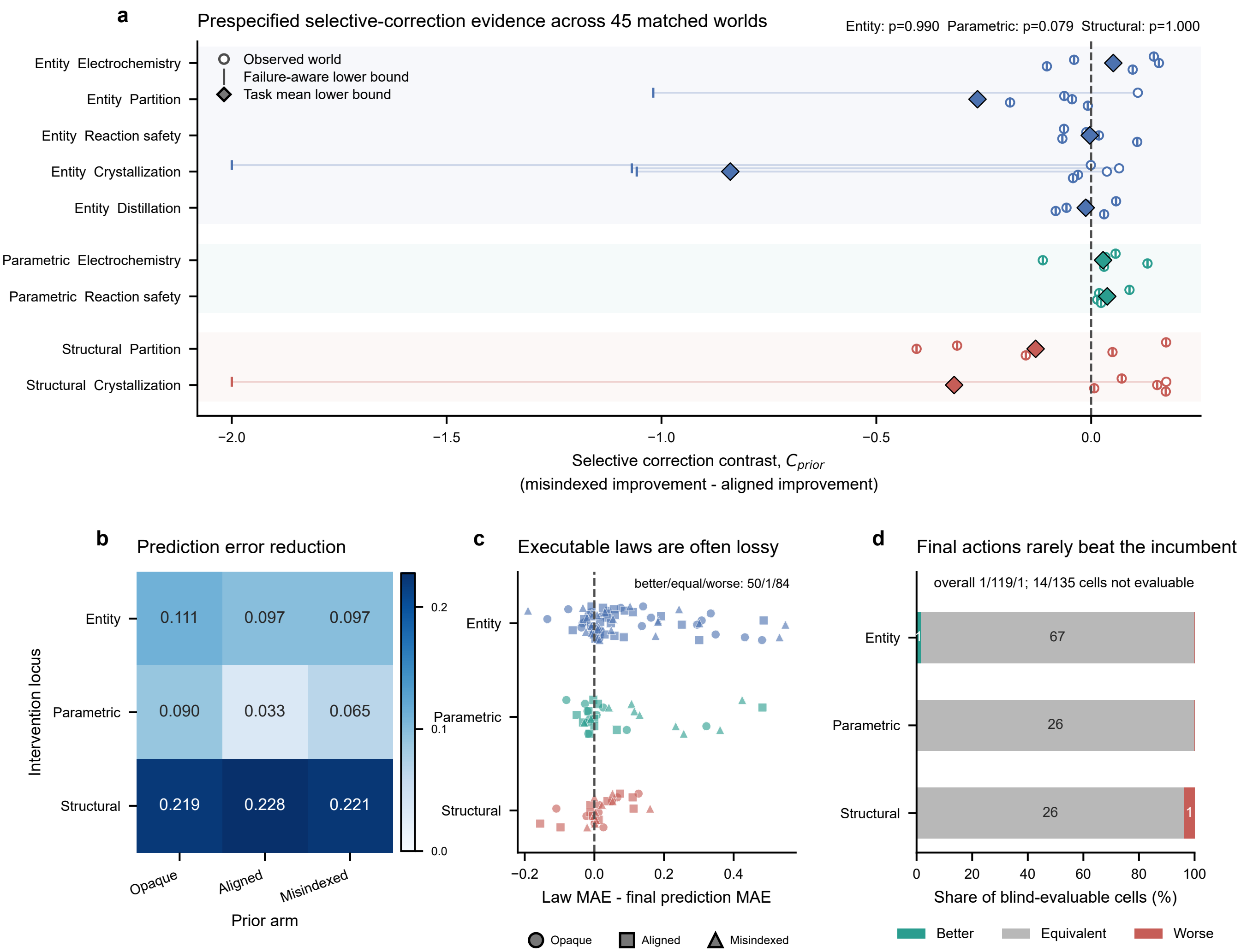

Figure 5. Predictive learning, selective correction, executable-law compression and incumbent replay dissociate. **a,** Failure-aware selective-correction contrasts across all 45 matched task–world clusters; circles show observed contrasts, vertical ticks retain adverse failure-aware bounds and diamonds show task means. Prespecified locus p values are shown. **b,** Mean pre-to-final normalized prediction-error reduction by intervention locus and initial-model arm. **c,** Final executable-law MAE minus final explicit-prediction MAE for all 135 cells. Positive values indicate lossy compression. **d,** Blind incumbent-replay outcomes for all 121 evaluable cells; 14 incomplete participant cells remain in the denominator as unstarted.

and incurred one resource rejection before adapting. Its final choice ranked 8/8 with normalized regret 1.0 and remained in the inadequate-law/wrong-action category. The repair shows that a fresh session can cross the original interruption while revealing a genuine resource-planning risk. It has a different trajectory and is not merged into the original 45-cell denominator.

7.4 The causal action-transfer follow-up failed before participant execution

The terminal matrix cannot identify whether acquired evidence caused better selection, so we prepared the separate five-condition follow-up described above. Its provider-free formal stage fixed 15 task–world clusters and 1,680 candidate, checkpoint and oracle-grid truth executions with exact replay. The first eight clusters completed **896/896** truth executions and exact replays without a provider call. All eight candidate-opportunity gates passed, and seven oracle laws passed the frozen rank criterion.

The eighth cluster was a fresh crystallization world. Its outcome-disjoint oracle law achieved Spearman rank correlation **0.738095** across the eight candidates, below the prespecified **0.80** threshold; its predicted Top-1 also disagreed with truth, while fit/candidate overlap remained zero. The formal preparation therefore stopped by design. The remaining seven clusters, operational canary, all 225 participant sessions and all 540 planned participant experiments were not started, and no world or unfavorable result was replaced. This result does not estimate a poor participant effect. It establishes that the current oracle-law control was not robust enough in fresh formal worlds to support the intended causal decomposition.

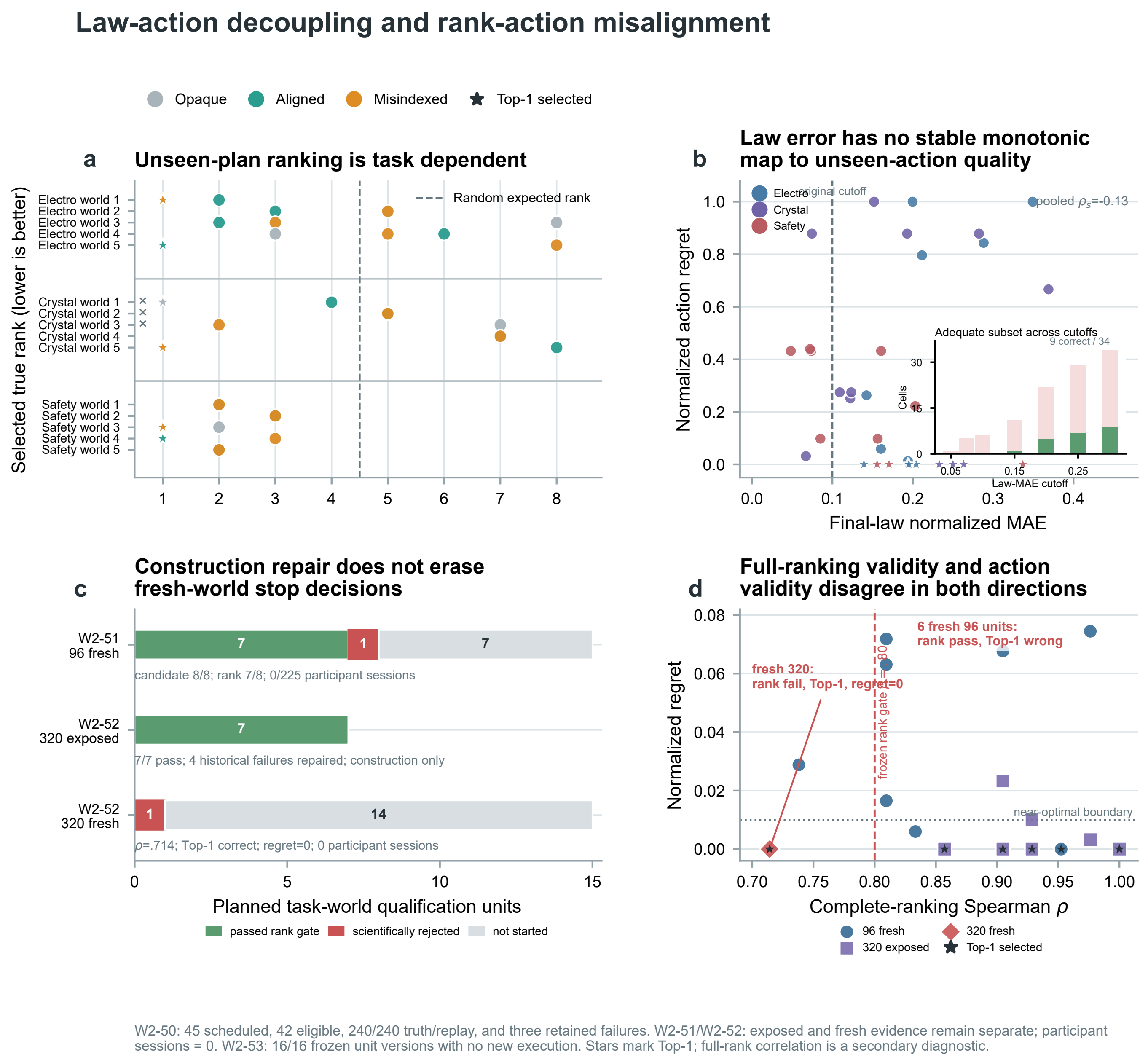

Figure 6. Agent law-action decoupling and evaluator rank-action misalignment. a, Selected true rank for 42 eligible W2-50 cells; three crystallization failures remain in the 45-cell scheduled denominator and the random-rank line is not an experimental control. **b,** Continuous final-law error versus normalized regret for the same cells. Stars mark Top-1 selections; colors identify tasks. The inset sweeps seven law-adequacy cutoffs and stacks correct and wrong actions within each adequate subset. **c,** W2-51 and W2-52 qualification funnels retain passed, scientifically rejected and unstarted units; exposed construction and fresh qualification are separate evidence roles, and participant execution is zero. **d,** Complete-ranking Spearman correlation and normalized regret for all 16 frozen W2-53 unit versions. Stars mark Top-1 selection; the panel adds no execution and changes no historical stop decision.

7.5 A denser oracle grid repairs exposed failures but not prospective ranking

We next tested whether the failure arose from sparse coverage rather than typed-law representation. The oracle grid was expanded from 96 queries to **320**: 64 global queries plus 256 queries around the candidate neighborhood. The fitted ExtraTrees family and the frozen Spearman threshold were unchanged, so the intervention isolated coverage. After a platform-defective partial run was retained and the evidence-reference limit repaired, the complete construction screen passed **7/7** exposed units with **2,352/2,352** truth executions and exact replays. All four historical failures were repaired and the minimum construction Spearman correlation was 0.857143.

Construction success did not transfer to the prospective block. Its first new electrochemical world completed **336/336** truth executions and exact replays but obtained Spearman correlation **0.714286**, below the frozen 0.80

gate. The oracle nevertheless selected the true Top-1 action with zero regret; fit/candidate overlap remained zero and candidate outcomes were not used for fitting. The stop rule therefore left the remaining 14 prospective clusters unstarted. Increasing coverage repaired known worlds but did not establish fresh-world complete-ranking validity.

7.6 Evaluator-level ranking and decision quality are different estimands

A frozen retrospective diagnostic compared the original ranking decision with action endpoints for all **16/16** completed oracle unit versions from the 96- and 320-query studies. Original Spearman and Top-1 values were reproduced exactly without new truth execution. Among the eight fresh 96-query formal preparation units, **7/8** passed the rank gate but only **1/8** selected the true Top-1 action and **3/8** selected within 0.01 of the optimum; six units were rank-pass/action-wrong. Conversely, the first fresh 320-query unit was rank-fail/action-correct with normalized regret zero.

The disagreement is bidirectional. Complete-ranking correlation measures global ordering, whereas the causal study needs a positive control for the selected action and its decision loss. The original gate and stop decisions remain valid for their frozen protocol, but the diagnostic shows that future control qualification must prospectively prioritize regret, near-optimal selection and near-tie-aware ordering, with full-ranking correlation retained as a secondary diagnostic.

8. From endpoint scores to a transition map of scientific agency

The experiments replace a scalar notion of scientific success with a transition map. The controlled intervention is the correctness of the agent's initial model. Downstream, the study observes whether that perturbation changes search, whether acquired evidence changes counterfactual predictions, whether those predictions can be compressed into an executable relation and whether the final state supports selection among unseen plans. A failure at one transition need not erase competence at another.

For the evaluated agent system, the map contains four major breaks. Initial models redirect search but do not selectively repair the wrong starting model. Direct matched evidence can produce near-exact numerical revision without structural-law recovery. Executable summaries can lose information present in explicit predictions. The fourth break is dual: at the agent level, thresholded law adequacy does not determine unseen-plan selection; at the evaluator level, complete-ranking validity does not determine action validity. These are not versions of the same score; they are empirically separable transformations.

The map also defines the next experiments without making them part of the current result. Context-reset artifact portability is required to test whether a learned representation survives outside the source conversation. We attempted to add no-evidence, yoked-evidence and artifact-only ranking controls for unseen-plan selection, but the oracle-law control did not qualify in a fresh formal world. Any future causal action-transfer study therefore requires a new control construction that first demonstrates robust fresh-world decision validity under action-aligned endpoints. Private within-family replication and matched cross-system studies would test stability and generality. Each is a distinct estimand with its own denominator.

9. Discussion

9.1 The scientific object is a transformation, not an endpoint

Endpoint success answers whether a trajectory found something useful. It does not identify why. Entity-level partition, structural crystallization and structural partition demonstrate three different routes to a good result: durable utility from correct information, a head start later narrowed by exploration, and search scaffolding supplied by both correct and incorrect structure. Even a wrong model may improve an endpoint by redirecting exploration. Endpoint score is therefore a property of an induced trajectory, not a direct measure of the truth of the agent's internal model.

The initial-prior intervention supplies the missing causal comparison. Because the world, public contract, tools, resources and task identity are fixed within matched clusters, differences among arms can be attributed to the starting epistemic condition under the evaluated agent configuration. That claim ends at the intervention. Prediction, law and action results locate observed downstream dissociations; they are not causal mediation estimates of how one hidden internal state produced the next.

9.2 Matched evidence separates acquisition, revision and identification

The most informative result is not simply that selective correction failed. It is where the failure moves when evidence is controlled. In free discovery, the agent may fail because it never acquires the decisive observation. Parametric matched evidence shows that this matters: all five misspecified agents rejected the supplied direction once the peak-and-collapse response was placed in front of them. Yet the corrected structural study goes further. Identical direct evidence drove numerical prediction error near zero in every arm, while none of the five misspecified summaries recovered the governing 1.75-power relation.

Accurate interpolation after diagnostic evidence is thus not equivalent to structural identification. An agent can revise numbers, adopt a useful empirical saturation description and still fail to express the relation that generated the evidence. At this five-world scale, the pattern resembles phenomenological interpolation rather than mechanistic identification: analogous to a Ptolemaic epicycle tracking an observed trajectory without expressing its governing dynamics, the saturation and endpoint summaries fit the supplied region while omitting the registered 1.75-power relation. This is an analogy, not an additional empirical claim. The distinction matters for scientific use because structural representations support counterfactual reasoning, communication and reuse in ways that a locally fitted endpoint model may not. The experiment does not establish a population rate of mechanism recovery; it demonstrates an identifiable transition that larger studies can target.

9.3 Understanding and action decouple at agent and evaluator levels

The executable-law interface reveals another conversion loss. Every submitted law executed, but 84 of 135 were less accurate than the corresponding final explicit predictions. Formal validity is therefore not faithful scientific compression. The agent may hold a conditional collection of predictions that cannot be preserved in one compact submitted relation.

At the agent level, the unseen-plan assay exposes a different boundary. Complete public ActionPlans remove hidden workflow defaults as an explanation, yet selection remains task dependent and thresholded law adequacy does not map monotonically to correct action. Because the study lacks a pre-exploration or no-evidence ranking baseline, this is not evidence for or against a causal action-transfer effect. It is evidence about the terminal state reached by the full campaign: some unseen plans are ranked well, but the ability is uneven and does not reduce to the submitted law.

At the evaluator level, the failed five-condition preparation sets a complementary boundary. A nominally oracle artifact is not a valid positive control merely because its fitting procedure is outcome-disjoint or succeeds in development worlds; it must preserve the decision-relevant ordering in the fresh worlds where the causal comparison is made. Stopping before participant execution prevents a weak control from turning an uninterpretable contrast into an apparent action-transfer effect.

The 96- versus 320-query diagnostic sharpens this point. Global ordering and decision quality can disagree in both directions: most fresh 96-query units passed the correlation gate while selecting the wrong action, whereas the first fresh 320-query unit failed the gate while selecting the optimum. Evaluator validity is therefore part of scientific-agent validity. A control should be qualified on the estimand needed by the causal contrast, not on a convenient surrogate that can succeed or fail for decision-irrelevant parts of the ranking.

9.4 The harness is part of the scientific system

Persistent sessions provide memory, repeated tool choice and belief revision that stateless calls do not. They also create failure surfaces through schema conformance, context growth, recovery rules and finite resources. All checkpoints were eventually recovered, but the large number of rejected typed submissions shows that the measurement interface affects the trajectory being measured. The participant is therefore the fixed combination of model, reasoning setting, prompt, persistent session, tools and resource policy. Cross-system comparisons must match or explicitly manipulate those components rather than attributing the result to model weights alone.

9.5 Scope and limitations

ChemWorld provides bounded executable causal worlds, not universal chemical fidelity or direct wet-laboratory validation. One fixed DeepSeek-V4-Flash agent configuration supports conclusions about that system, not language models in general. The prospective programme spans nine task–locus combinations but only five independent worlds per task. Observation-model interventions, private confirmation, cross-system replication and context-reset artifact portability were not executed.

The corrected structural matched-evidence result contains five worlds and is explicitly non-confirmatory. The earlier structural run affected by an evaluator omission is excluded. The unseen-plan study has 42 eligible cells, three retained crystallization failures and only 12 complete three-arm clusters; arm means are descriptive. Its random-rank line is not an experimental control. The separate five-condition follow-up produced no participant data: one of the first eight formal oracle controls failed its frozen rank criterion, and the remaining seven clusters were not started. The denser oracle construction passed only on seven exposed units and its prospective qualification stopped after the first new world; the rank/action alignment result is a retrospective diagnostic over 16 completed unit versions, not a new causal participant study. The single-stratum matched extension pilot is development-only and does not repair the missing five-condition estimate. Ten prospective cells also contain discard-affected checkpoint timing that cannot be repaired retrospectively. Exact software replay preserves execution semantics but does not erase interface burden or implementation variability.

9.6 Conclusion

The question is not whether an AI scientist finds a good experiment, but what that experiment changes in its model of the world. Controlled interventions on the starting model, evaluator-owned counterfactuals and matched evidence make those changes separately observable. In the present agent, search, prediction, selective correction, structural identification, executable compression and terminal decision do not rise and fall together, and the evaluator's complete-ranking gate does not substitute for decision validity. Scientific agency is not a score; it is a sequence of evidence-driven transformations. The final break has two distinct locations: understanding can decouple from action inside the agent, and the evaluator's ranking surrogate can decouple from the action validity it is meant to certify.

10. Methods

10.1 World and initial-model construction

Each task instantiates an executable $W = (\mathcal{E}, G, \Theta, O, C)$ and a participant-facing $M_0 = (\widehat{\mathcal{E}}, \widehat{G}, \widehat{\Theta}, \widehat{O}, \widehat{S})$. Prospective worlds are selected deterministically from a set disjoint from exploratory worlds. Within each world cluster, all arms share W, the resource card and stochastic identity; exactly one declared component of M_0 changes. In the entity locus, aligned and misindexed dossiers contain identical fields, values, wording and confidence language, while the latter applies a prespecified permutation to material identifiers. Structural, parametric and observation-model extensions alter only their declared agent-facing representation while retaining the external world and contract. A layer extension is included only after a separate identifiability analysis confirms that aligned and misspecified encodings are matched in information volume, wording, confidence, baseline plausibility and falsification cost.

10.2 Transactional execution and resources

Every operation enters schema validation and resource preflight before candidate execution. Committed operations update physical state and the campaign ledger; invalid or resource-rejected attempts retain their declared reporting debit without entering committed physical state. Task-specific resource cards bound vessel starts, assays, measurements, stocks, process time, repeated operations, quench and transfer time, and final-assay reserve across all experiments in a campaign.

10.3 Persistent agent and tool execution

Each participant cell launches one persistent agent process and retains one session across the complete pattern-owned campaign. The participant instructions prohibit shell use, file changes and repository inspection and require physical decisions to pass through the host-owned laboratory interface. The bounded domain tools expose material information, belief checkpoints, operation submission, public state and history, artifact inspection and final recommendation commitment. The host validates and executes submitted actions but never chooses a fallback scientific action.

Every operation submission contains a brief structured rationale stating its expected effect, diagnostic target and evidence dependence. Tool records retain call order, status, timestamps and error classes without retaining raw interaction payloads or private chain-of-thought. An infrastructure retry or resume is an operational attempt within the same cell, not a new experiment or independent sample. Belief checkpoints are tool calls inside the existing session rather than separate conversations.

10.4 Belief and law-summary checkpoints

Checkpoint records contain prior assessment, predictions, uncertainty, evidence references, an executable law summary, next-experiment intent and overall confidence. The schema permits bounded rationales but not an unconstrained persistent notebook. After the campaign, explicit predictions and the final law summary are evaluated against sealed truth packs. The summary must execute for the exact prespecified query-metric set; the analysis records its normalized error, pre-to-summary improvement, error relative to the effective final checkpoint and prediction-consistency error. These quantities are reported continuously. They are not converted post hoc into a public binary law-discovery label, and a reusable or transferable-law claim additionally requires independent transfer evidence.

The schema-capacity control refits every complete final prediction vector with legal identity-link laws over the registered feature coordinates and allowed bases. The full fit permits up to 64 terms per metric; the term-matched fit uses the participant's submitted per-metric term budget; and the leave-one-query-out fit withholds each registered query in turn. Every fitted payload must pass the production parser and executor, and executor predictions must agree with the independently computed design-matrix predictions within 10^{-10}. These are same-domain representation and distillation controls, not tests of global mechanism recovery or transfer.

10.5 Statistical analysis

The prospective cohort contains 45 independent matched task–world clusters: 25 entity-level, ten parametric and ten structural. Every cluster contains three participant arms, which are paired interventions rather than independent samples. Prediction-to-law inference is performed separately by locus. The entity locus uses a three-component failure-aware intersection–union criterion; the parametric and structural loci use task-fixed-effect contrasts, require both task means to be positive and retain adverse bounds for failed or unscorable arms. A global cross-locus decision requires all three locus criteria to pass, and naive pooling across the nine task–locus combinations is forbidden. Endpoint contrasts in Section 6 are descriptive trajectory outcomes, not substitutes for that prespecified prediction-error analysis. Prespecified sensitivity analyses include observed-point, complete-case, heteroscedasticity-robust and task-stratified cluster-bootstrap summaries.

Matched-evidence analyses use the world as the inference unit and retain the prespecified contrast $(E_{mis,pre} - E_{mis,post}) - (E_{aligned,pre} - E_{aligned,post})$. The structural matched-evidence study uses five worlds, so all 2^5 sign flips are enumerated for the exact one-sided directional check and a Student-t interval is reported descriptively. Protocol-validation sessions are excluded. Structural-recovery counts use only the submitted model summary and evidence assessment, never private reasoning text.

10.6 Open-action plan construction and analysis

For the formal multi-task action assay, an outcome-blind generator constructed eight candidate ActionPlans per world across three task families and five worlds per task before participant execution. Candidate selection used only the declared plan coverage criteria and could not read evaluator scores, participant trajectories, checkpoint outcomes or later model responses. Every plan began from the same fresh-batch contract and disclosed its full operation sequence and parameters. Independent validation executed all candidate and checkpoint truth queries without additional model calls, verified exact replay and confirmed that the disclosed plan was the plan evaluated.

After 12 participant experiments and the final checkpoint, the participant returned a complete ranking and selected one candidate. Evaluator-owned outcomes were then used to calculate selected rank, Top-1, raw regret, normalized regret, complete-ranking agreement and the joint mechanism–action category. A law was classified as adequate when its normalized held-out mechanism MAE was at most 0.05 under the frozen evaluator threshold; otherwise it was classified as inadequate. This binary label was fixed before action outcomes were interpreted. A cell was eligible only when all 12 experiments, five checkpoints, final ranking, resource reconciliation and execution-integrity checks completed without contamination. All 45 scheduled cells remained in the denominator. Arm summaries are descriptive because three crystallization cells were ineligible and only 12 task–world clusters retain all three arms. The three additional interface validations used one world seed per task and were evaluated only as interface checks, not as multi-world scientific evidence.

The continuous law–action control retained the same 42 eligible cells and never imputed the three historical failures. Pearson and Spearman associations were calculated pooled and task-stratified. Intervals resampled the 15 frozen task–world clusters, retaining every eligible arm in a sampled cluster, for 10,000 bootstrap replicates with seed 20260827. Threshold sensitivity used the seven fixed law-MAE cutoffs 0.05, 0.075, 0.10, 0.15, 0.20, 0.25 and 0.30; action outcomes did not select cells, tasks, bases or cutoffs.

The five-condition causal follow-up paired each autonomous donor with no-evidence, yoked-evidence, learned-law-only and oracle-law recipients within task–world–prior strata. Donor failure would have retained the donor and marked its yoked and learned-law descendants as not started; donor replacement was forbidden. The oracle law used the same typed schema and scientific scope as the learned law, but was fitted provider-free on 96 registered queries disjoint from the eight candidates. Formal expansion required every task–world oracle to reach candidate-order Spearman correlation at least 0.80. The formal stage completed 896 truth executions and exact replays across eight clusters before a crystallization oracle obtained 0.738095 and rejected the block. Because no participant session was started, none of the prespecified autonomous-minus-no-evidence, yoked-minus-no-evidence, autonomous-minus-yoked, learned-law-minus-no-evidence or oracle-minus-learned-law contrasts was estimated.

The independent large-grid study increased oracle coverage to 64 global and 256 candidate-neighborhood queries while holding the fitted ExtraTrees family fixed. Construction used seven exposed unit versions; prospective qualification used new worlds and the same frozen Spearman threshold. The first new world failed at 0.714286 after 336 truth executions and exact replays, despite Top-1 agreement and zero regret, so 14 planned worlds remained unstarted. A separate retrospective alignment analysis re-read the 16 completed 96- and 320-query unit versions, reproduced their original Spearman and Top-1 outcomes, and added no execution. Top-1, regret, selection within 0.01 of the optimum and near-tie-aware pair ordering were treated as action endpoints; no result was used to revise an earlier gate or stop rule.

10.7 Optional private confirmation boundary

No private participant cohort was executed for the present study. If private confirmation is pursued, it will use newly sealed world instances disjoint from exploratory and public worlds, retain the same three-arm participant and evaluator contracts, and preserve every completed, failed and unstarted cell in a one-shot denominator. Such a cohort would test within-family replication. A separate context-reset artifact-portability design would still be required for a compositional-transfer claim.

10.8 Reproducibility and failure accounting

Participant trajectories, evaluator truth sets and blind-replay sets are stored separately and joined through stable record identifiers. Every completed participant trajectory must pass physical replay, campaign-resource replay and hidden-boundary verification. Process attempts, sessions, tool calls, operation attempts, committed operations, complete experiments, cells and evaluator executions are reported with distinct denominators.

11. Data and code availability

The current source release contains the executable environment, prespecified protocols, analysis code, source data and reproducible figure scripts used for this manuscript. The fixed agent-system configuration, prompt contract, reasoning setting and sampling parameters are bound in the release metadata. Raw interaction payloads and credentials are excluded, and no private cohort data are included because that optional study was not executed.

12. Competing interests

The authors declare no competing interests.

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